
The Frontier of LLM Evaluation: Item Response Theory

About Me

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ChiKhaPo: A Large-Scale Multilingual Benchmark for Evaluating Lexical Comprehension and Generation in Large Language Models

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Task	Comprehension: $X \rightarrow \text{model}$	Generation: $\text{model} \rightarrow X$
Word Translation	Input: Translate the following text from Malay to English: ujan. Correct Output: rain Model Output: rain Score: scores["ujan"] += 1	Input: Translate the following text from English to Afrikaans: attacked. Correct Output: aangeval Model Output: aangeval Score: scores["aangeval"] += 1
Word Translation with Context	Input: In 'Minonke konam phoro isi sonturi aghaipo aro anang pen Jisu yok honsi kido, aro alok hel, labadi chiklik hel aro ajat jat kachiplang theksi, anali chiphere detno, aro pulo, "Khanangsi labang arlengpo Arnam Aso kido . ' ' ', the word 'kido' means ____ in English. Correct Output: letter Model Output: child Score: scores["kido"] += 0	Input: In 'After the match, King of Clay said, "I am just excited about being back in the final rounds of the most important events. I am here to try to win this." ', the word 'win' means ____ in Basque. Correct Output: aurrea hartu Model Output: ganar Score: scores["aurrea hartu"] += 0
Translation-Conditioned Language Modeling	Input: Translate the following text into English: Dyula: Aka dugutaga se'n fei, Iwasaki ye kassara chaman le sôrô. English: During his trip, Iwasaki Reference Translation: During his trip, Iwasaki ran into trouble on many occasions. Model Output: $P[\text{ran} \mid \text{input}] = 0.567$ Score: scores["kassara"] += 0.567	Input: Translate the following text into Iloko. English: "We now have 4-month-old mice that are non-diabetic that used to be diabetic," he added. Iloko: "Addaan kami ti 4-a-bulan a Reference Translation: "Addaan kami ti 4-a-bulan a babbao a dati ket diabetic ngem saan itan," nainayonna. Model Output: $P[\text{babbao} \mid \text{input}] = 0.351$ Score: scores["babbao"] += 0.351
Bag-of-Words Machine Translation	Input: Translate into English: Los trabajadores devon sovent obtenir l'aprobacion de sos superiors. Reference Translation: Workers must	Input: Workplace harmony is crucial Reference Translation: Ukusebenza

Ruder (2021). Challenges and Opportunities in NLP Benchmarking.

Ruder (2024). The Evolving Landscape of LLM Evaluation.

Evaluation

Why is model evaluation **important**?

How are researchers **disadvantaged** with current day evaluation methods?

Human evaluation is the gold standard in many NLP tasks but it suffers from **prohibitive costs** and **poor scalability** (Zouhar et al, 2026).

In the absence of human supervision, researchers rely on [**benchmarking** and] automated metrics (Zouhar et al, 2026).

*“Datasets are the
telescopes of our field.”*

– Aravind Joshi

benchmark

[noun]

A standard point of reference against which things can be compared. Consists of dataset(s), metric(s), and a way to aggregate performance.

Recent models have “**outpaced** the benchmarks [that] test them” (AI Index Report, 2021).



Benchmark saturation over time for popular benchmarks. Initial performance and human performance are normalized to -1 and 0 respectively (Kielbaso et al, 2021).

Evaluating generative AI models is a routine, but **resource-intensive**, process that is conducted over and over again during the course of model development (Fisch et al, 2026).

Multilingual models are evaluated only on a small subset of languages. There is no guarantee that performance on English-Spanish, the only common evaluation dataset, is representative of the expected performance of the models over all other language pairs (Anastasopoulos and Neubig, 2020).

(Efficient?) Evaluation

What makes these evaluation methods **efficient**?

Why should researchers be **skeptical**?

Using historical data

“The systems under comparison often share architecture, training data, or system prompts so their performances are correlated” (Fisch et al, 2026).

Literature Review

CollabEval (Fisch et al, 2026) reframes model evaluation as a **matrix completion problem**, using results from other historical models to impute scores for skipped evaluation prompts from a potentially much larger evaluation set.

NLPerf (Xia et al, 2020) developed a **regression model** to predict the performance on a particular language and experimental setting given **past experimental records** as well as **dataset, language, training, and model architectural** features.

Adaptive Transfer Scaling Law or **ATLAS** (Longpre et al, 2025) is composed of (i) a cross-lingual matrix spanning 1444 language pairs and (ii) **a language agnostic scaling law** that reveals how to optimally scale model size and data when adding languages without sacrificing performance.

Montgomery et al (2025) developed a framework that jointly models downstream performance as a function of the **training compute and provided context**

Microbenchmark: Shrinking the benchmark

The dominant practice of relying on aggregate metrics such as mean accuracy is **deceptively simple**, treating all items as equally informative regardless of their difficulty or discriminative power (Ethayarajh and Jurafsky, 2020; Rodriguez et al, 2021). This simplification **dilutes informative signals in evaluation data, renders model rankings sensitive to benchmark composition, and hinders the accumulation of comparable findings** (Jiang et al, 2026; Gill et al, 2025).

Literature Review

AnchorPoints (Vivek et al, 2024) reduces selecting a smaller representative benchmark to solving a K-Medoids problem that maximizes the correlation between selected and remaining points. Each point is represented by model confidences.

tinyBenchmarks (Polo et al, 2024) uses **Item Response Theory** to learn representations of benchmark examples and clustering to find a more robust evaluation set. Researchers conclude that 100 curated benchmark examples are enough to reliably estimate the performance of various LLMs within $\approx 2\%$ error.

Fluid Benchmarking (Hofmann et al, 2025) estimates an **item response** model based on existing LM evaluation results and uses the inferred quantities to select evaluation items **dynamically**.

Arviv et al (2026) call for the adoption of **sequential testing**—a statistical method that allows you to evaluate data as it is collected instead of waiting until a test reaches a fixed size—with stopping criteria (e.g. diminishing returns detection)

Item Response Theory

What **is** Item Response Theory?

What are its **assumptions**?

What happens if those
assumptions are **violated**?



Subjects/Examinees

(Test) items

Consider these two new statistics that we have not yet studied:

- $\text{mid-range} = (\text{minimum} + \text{maximum}) / 2$
- $\text{mid-hinge} = (\text{lower quartile} + \text{upper quartile}) / 2$

a) Is the mid-range a measure of center or a measure of variability?

- A. Center
- B. Variability

b) Is the mid-hinge a measure of center or a measure of variability?

- A. Center
- B. Variability

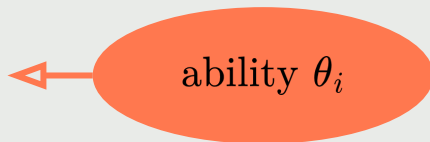
c) Is the mid-range resistant to outliers?

- A. Yes
- B. No

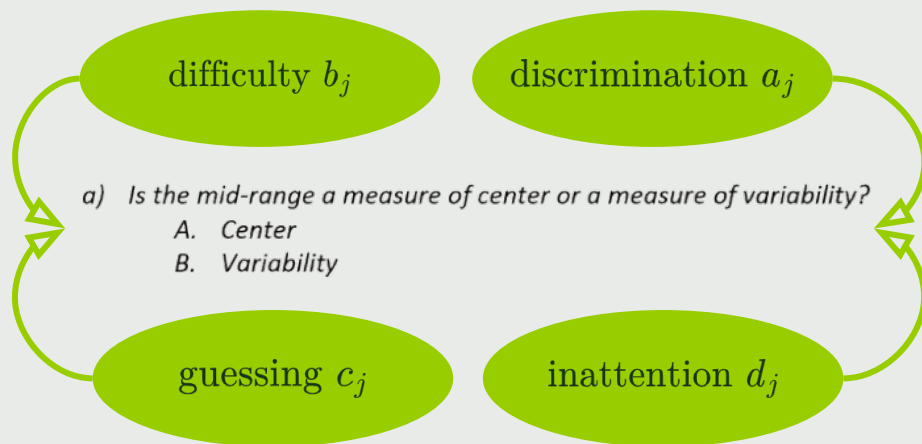
d) Is the mid-hinge resistant to outliers?

- A. Yes
- B. No





A subject i is characterized by its ability θ_i



An item j is characterized by its difficulty b_j , discrimination a_j , guessing c_j , inattention d_j

Dichotomous

Dichotomous IRT models have two possible item scores

True	<i>Yes</i>
False	<i>No</i>

Please select one of the following

- Choice A
- Choice B
- Choice C
- Choice D

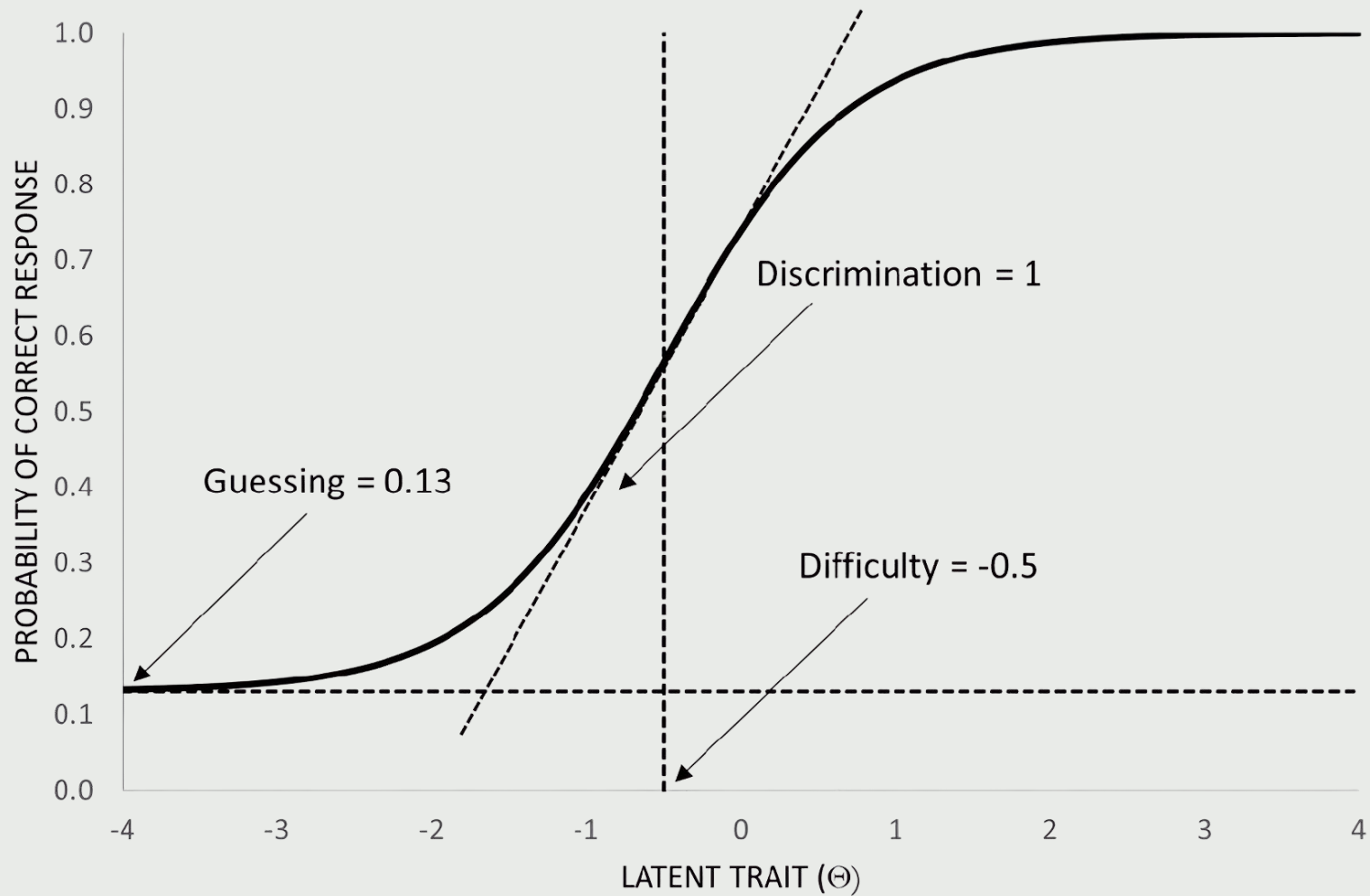
Polytomous

Polytomous IRT models have more than two possible scores

Please rate your discomfort on a scale of 1 to 5:

1 2 3 4 5

Item Response Theory explains the **relationship between an individual's ability and their observed performance** (Columbia University Mailman School of Public Health). Formally, the probability of correctly answering an item is a **function of the person's ability** and visualized as the **item response curve**.



$$1 \text{ PL} \quad \mathbb{P}[\mathbf{r}_{ij} = 1 \mid \theta_i, b_j] = \frac{\exp(D(\theta - b_j))}{1 + \exp(Da(\theta - b_j))} = \frac{1}{1 + \exp[-Da(\theta_i - b_j)]}$$

$$2 \text{ PL} \quad \mathbb{P}[\mathbf{r}_{ij} = 1 \mid \theta_i, a_j, b_j] = \frac{\exp(\theta_i - b_j)}{1 + \exp[a_j(\theta_i - b_j)]} = \frac{1}{1 + \exp[-Da_j(\theta_i - b_j)]}$$

$$3 \text{ PL} \quad \mathbb{P}[\mathbf{r}_{ij} = 1 \mid \theta_i, a_j, b_j, c_j] = c_j + (1 - c_j) \frac{1}{1 + \exp[-Da_j(\theta_i - b_j)]}$$

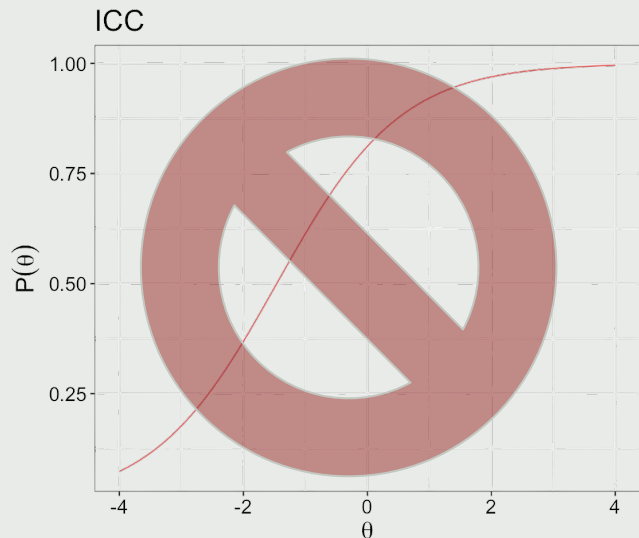
$$4 \text{ PL} \quad \mathbb{P}[\mathbf{r}_{ij} = 1 \mid \theta_i, a_j, b_j, c_j, d_j] = c_j + (d_j - c_j) \frac{1}{1 + \exp[-Da_j(\theta_i - b_j)]}$$

Assumptions

1. **Monotonicity:** as the trait level “increases,” the probability of a correct response increases.
2. **Unidimensionality:** there is one dominant latent trait being measured, and this trait is the driving force for the responses observed for each item in the measure.
3. **Local independence:** responses given to separate items in a test are mutually independent given a certain level of ability. A correct/incorrect response to one item should not lead to a correct/incorrect response to another.
4. **Invariance:** we can estimate item parameters from any position on the item response curve. Put another way, invariance means that item parameters do not depend on the characteristics of a particular population.

VOTE: Which of these assumptions would be the most problematic if violated?

What if monotonicity fails?



$$1 \text{ PL} \quad \mathbb{P}[\mathbf{r}_{ij} = 1 | \theta_i, b_j] = \frac{1}{1 + \exp[-Da(\theta_i - b_j)]}$$

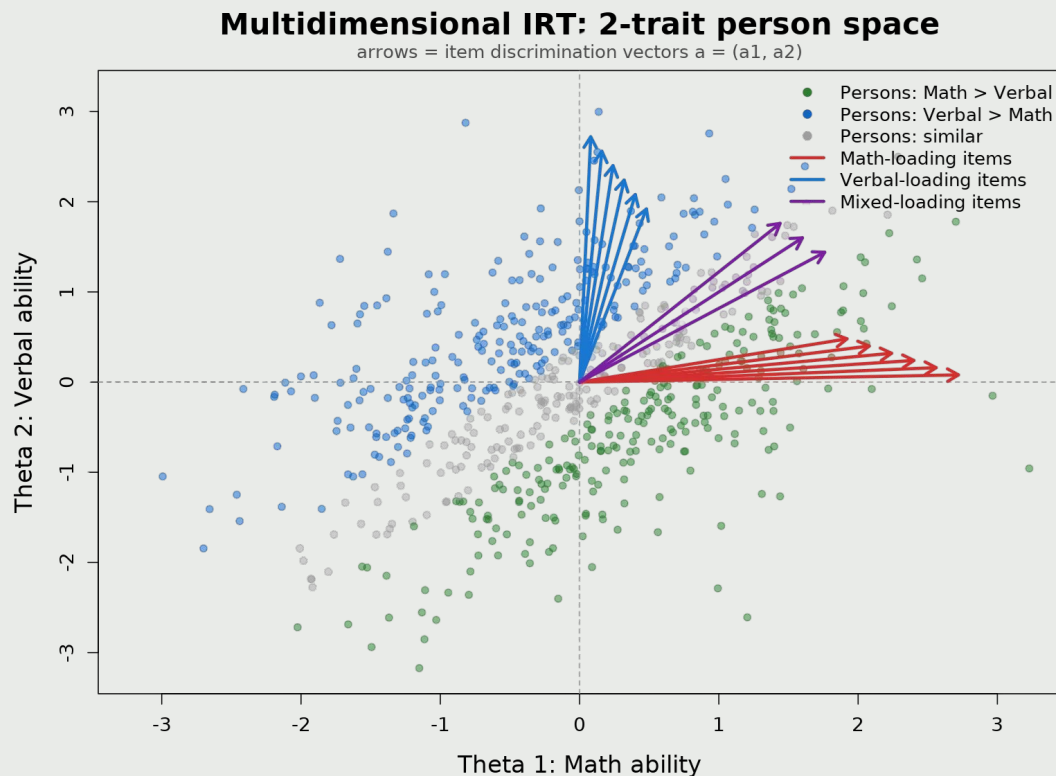
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$$3 \text{ PL} \quad \mathbb{P}[\mathbf{r}_{ij} = 1 | \theta_i, a_j, b_j, c_j] = c_j + (1 - c_j) \frac{1}{1 + \exp[-Da_j(\theta_i - b_j)]}$$

$$4 \text{ PL} \quad \mathbb{P}[\mathbf{r}_{ij} = 1 | \theta_i, a_j, b_j, c_j, d_j] = c_j + (d_j - c_j) \frac{1}{1 + \exp[-Da_j(\theta_i - b_j)]}$$

Standard parametric IRT models fail to accurately capture the relationship between latent ability and the probability of getting a specific (item) response

What if unidimensionality fails?



Unidimensional
2PL IRT

$$\mathbb{P}[\mathbf{r}_{ij} = 1 \mid \theta_i, a_j, b_j] = \frac{1}{1 + \exp[-Da_j(\theta_i - b_j)]}$$

Multidimensional
2PL IRT

$$\mathbb{P}[\mathbf{r}_{ij} = 1 \mid \boldsymbol{\theta}_i, \mathbf{a}_j, \mathbf{b}_j] = \frac{1}{1 + \exp\left(-D \sum_{k=1}^K a_{jk}(\theta_{ik} - b_{jk})\right)}$$

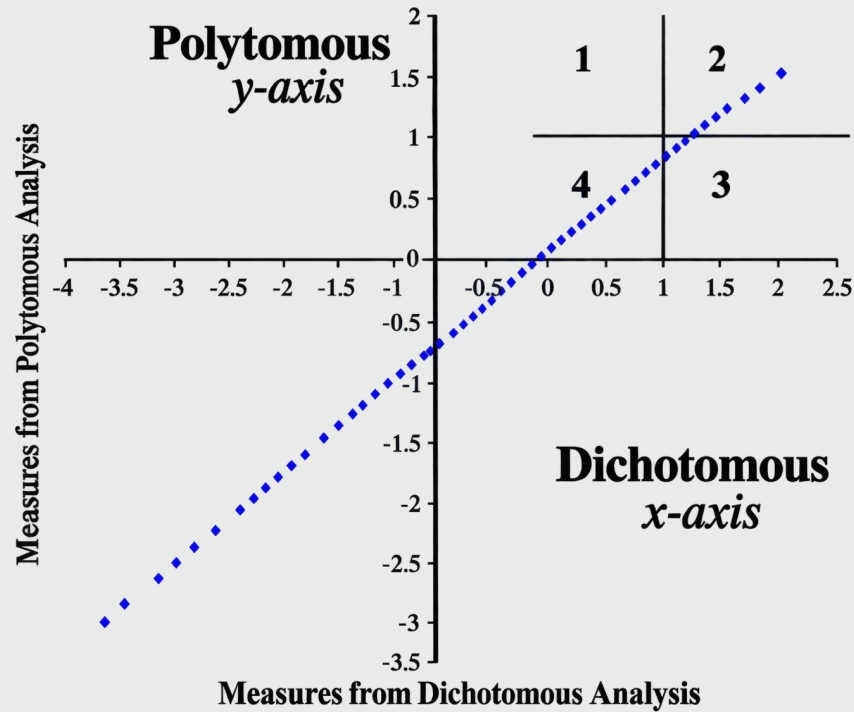
What if local independence fails?

$$\mathbb{P}[\mathbf{r}_{ij} = 1 \wedge \mathbf{r}_{ij'} = 1 \mid a_j, b_j, c_j, d_j, a_{j'}, b_{j'}, c_{j'}, d_{j'}, \theta_i] \neq \mathbb{P}[\mathbf{r}_{ij} = 1 \mid a_j, b_j, c_j, d_j, \theta_i] \times \mathbb{P}[\mathbf{r}_{ij'} = 1 \mid a_{j'}, b_{j'}, c_{j'}, d_{j'}, \theta_i]$$

*The antithesis of local independence is **local dependence***

Estimates of the latent variables and item parameters will be **incorrect**. Local dependence can **inflate reliability** and give a **false impression** of the precision and quality of the test.

Items are **redundant**. Test statistics such as X^2 and G^2 test statistics (Chen and Thissan, 1997), Yen's Q_3 (Christensen et al, 2016)



A set of locally dependent items can be bundled into a **polytomous** super-item to partial out the influence of local item dependence.

What if invariance fails?

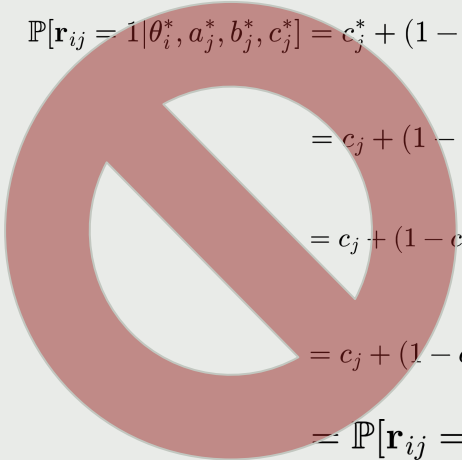
Because the latent (ability) scale in an IRT model is arbitrary, (unlinked) item and examinee parameters are identical only up to a set of linear transformations listed below (Lord, 1980; Rupp and Zumbo, 2006). Should invariance fail, **this equivalence fails to hold.**

$$\theta^* = A\theta + B$$

$$a^* = a/A$$

$$b^* = Ab + B$$

$$c^* = c$$



$$\begin{aligned}\mathbb{P}[\mathbf{r}_{ij} = 1 | \theta_i^*, a_j^*, b_j^*, c_j^*] &= c_j^* + (1 - c_j^*) \frac{1}{1 + \exp[-Da_j^*(\theta_i^* - b_j^*)]} \\ &= c_j + (1 - c_j) \frac{1}{1 + \exp\left[-D\frac{a_j}{A}(A(\theta_i + B) - (Ab_j + B))\right]} \\ &= c_j + (1 - c_j) \frac{1}{1 + \exp\left[-D\frac{a_j}{A}(\mathcal{A}(\theta_i + b_j))\right]} \\ &= c_j + (1 - c_j) \frac{1}{1 + \exp[-Da_j(\theta_i - b_j)]} \\ &= \mathbb{P}[\mathbf{r}_{ij} = 1 | \theta_i, a_j, b_j, c_j]\end{aligned}$$

What if **invariance** fails?

If parameters are not invariant, the statistical foundation for inference is not identical across populations and hence the inferences (from one population) are **not generalizable** across to the (other population) (Rupp and Zumbo, 2006).

A breakdown of invariance often signals **differential item functioning**, meaning members of specific demographic, cultural, or clinical subgroups have a different probability of endorsing an item even when they have the “exact same ability.”

One or more subpopulations of examinees may have an expected **advantage** over others, raising concerns about **fairness in assessment and accountability** (Huggins, 2012).

How do we estimate
 $\theta_i, a_j, b_j, c_j, d_j$?

Marginal Maximum Likelihood

MML assumes that abilities are sampled from some larger distributions denoted as $F(\Theta)$. Typically, IRT modelers assume that the distribution F is the normal distribution with mean 0 and standard deviation 1. However the normal distribution does not necessarily work for all applications.

$$(\hat{a}, \hat{b}, \hat{c}, \hat{d}) = \underset{a, b, c, d}{\operatorname{argmax}} \mathcal{L}(a, b, c, d \mid \mathbf{R}) = \prod_i \mathbb{P}[\mathbf{r}_i \mid a, b, c, d] = \prod_i \int_{\theta} L_i(\theta_i \mid \mathbf{r}_i, a, b, c, d) dF(\theta_i)$$

Location and scale constraints are required to achieve a unique solution. The constraints can either be placed on the mean and standard deviation of the ability distribution $F(\Theta)$ or on the item parameters.

Variational Inference

Wu et al. (2020) proposed Variational Item Response Theory Lower Bound (VIBO), a variational inference method for fitting IRT models to large datasets. They assume:

$$\theta_i \in \mathbb{R}^K : \mathbb{P}[\theta_i] = \prod_{k=1}^K \mathbb{P}[\theta_{i,k}] \text{ where } \mathbb{P}[\theta_{i,k}] \sim \mathcal{N}(0, 1)$$

$$\mathbb{P}[\{a_j, b_j, c_j, d_j\}_{j=1:M}] = \prod_{j=1}^M \mathbb{P}[\{a_j, b_j, c_j, d_j\}] \text{ where } \mathbb{P}[\{a_j, b_j, c_j, d_j\}] \sim \mathcal{N}(0, 1)$$

$$q_\phi(\{a_j, b_j, c_j, d_j\}_{j=1:M} \mid r_{i,1:M}) = q_\phi(\{a_j, b_j, c_j, d_j\}_{j=1:M}) = \prod_{j=1}^M q_\phi(\{a_j, b_j, c_j, d_j\}_{j=1:M})$$

Variational Inference

The goal of VIBO is to pick a family of distributions that "best approximates the true posterior by minimizing an estimate of mismatch between true and approximate distributions." Latent variable $z \in Z$ represents ability and item characteristics $\theta_i, \{a_j, b_j, c_j, d_j\}_{j=1:M}$.

$$q_{\psi^*(r_i)}(z) = \arg \min_{q_{\psi(r_i)}} D_{\text{KL}}(q_{\psi(r_i)}(z) \parallel p(z \mid r_i)) = \arg \max_{\psi(r_i)} \mathbb{E}_{q_{\psi(r_i)}(z)} \left[\log \frac{p(r_i, z)}{q_{\psi(r_i)}(z)} \right]$$

by definition: minimizing the KL-divergence to the log posterior is equivalent to maximizing the ELBO

Variational Inference

The average quality of the variational approximations is

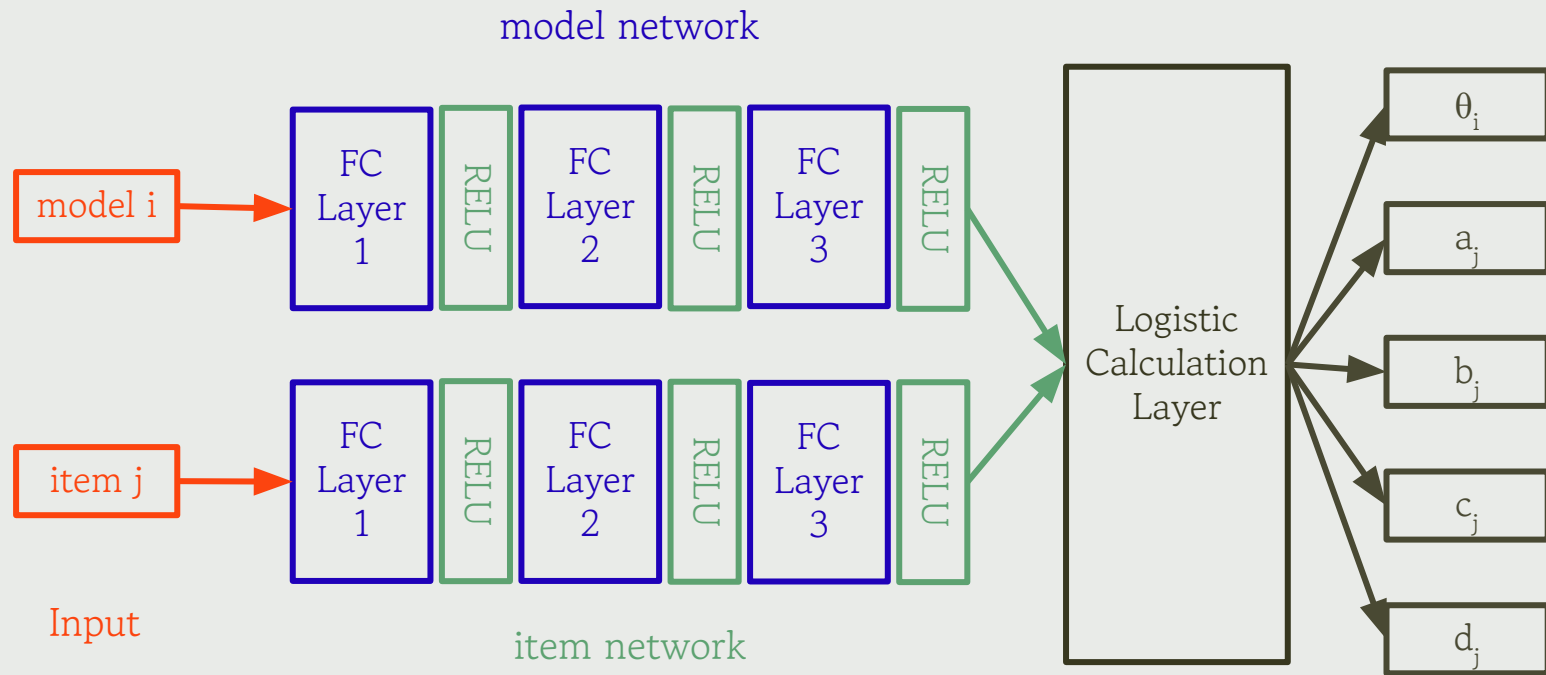
$$\mathbb{E}_{\mathbb{P}_D(r_i)} \left[\max_{\psi(r_i)} \mathbb{E}_{q_{\psi(r_i)}(z)} \left[\log \frac{\mathbb{P}[r_i, z]}{q_{\psi(r_i)}(z)} \right] \right] \quad \text{where } \mathbb{P}_D(r_i) \text{ is an empirical distribution over the observed variables}$$

Learning an approximate posterior for each $x \in D$ can grow to be unwieldy in a large dataset.

Wu et al (2020) proposed an amortized function f_ϕ (parameterized by ϕ) that maps a person's responses directly to parameters of their approximate posterior distribution. The number of parameters in amortization is vastly smaller than learning a per-observation posterior.

$$\max_{\phi} \mathbb{E}_{\mathbb{P}_D(r_i)} \left[\mathbb{E}_{q_\phi(z|r_i)} \left[\log \frac{\mathbb{P}[r_i, z]}{q_\phi(z | r_i)} \right] \right]$$

Neural Pseudo-Siamese Estimator



Neural Pseudo-Siamese Estimator

$$J(\theta) = -\frac{1}{\text{batch size}} \sum_{n=1}^{\text{batch size}} [y_n \log \mathbb{P}[r_n = 1 \mid a_{j_n}, b_{j_n}, c_{j_n}, d_{j_n}, \theta_{i_n}] + (1 - y_n) \log (1 - \mathbb{P}[r_n = 1 \mid a_{j_n}, b_{j_n}, c_{j_n}, d_{j_n}, \theta_{i_n}])]$$

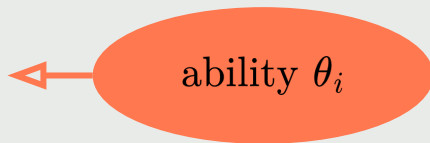
cross entropy loss function in Logistic Calculation Layer

The training objective is to minimize the discrepancy between the model's prediction and the observed binary outcomes. All learnable weights within both the model and item networks are calculated concurrently optimizing the estimated properties for both models and benchmark items.

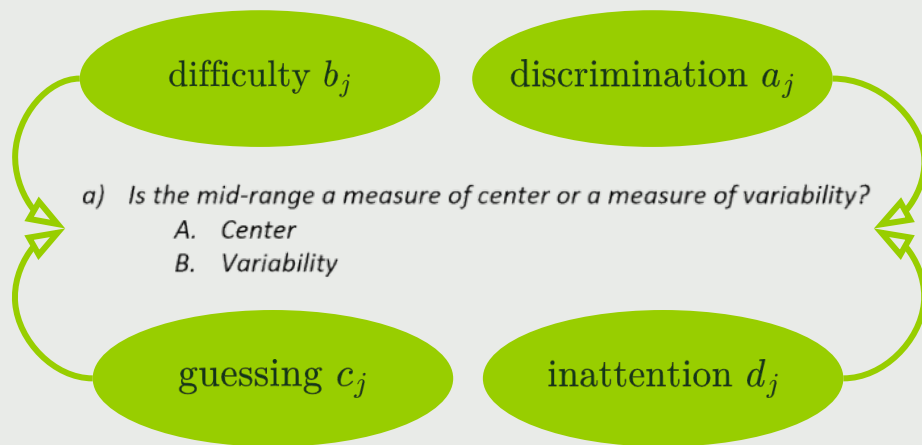
Method	Pros	Cons
Maximum Marginal Likelihood	The most conventional approach in human testing	• Computationally burdensome for large item benchmarks • Does not yield reliable estimates even when runs converged
Markov Chain Monte Carlo	Most consistent parameter recovery	Computational cost limited its feasibility for combinations of large benchmarks, large sample sizes, and more complex IRT models
Variational Inference	• Fast • Recovered model rankings slightly better	Unreliable sometimes for aggregate score recovery and item parameter recovery
Pseudo-Siamese IRT	• Computationally manageable • Showed competitive performance	

Arguments for (9/17) and against (9/18)
using Item Response Theory in NLP

Recap

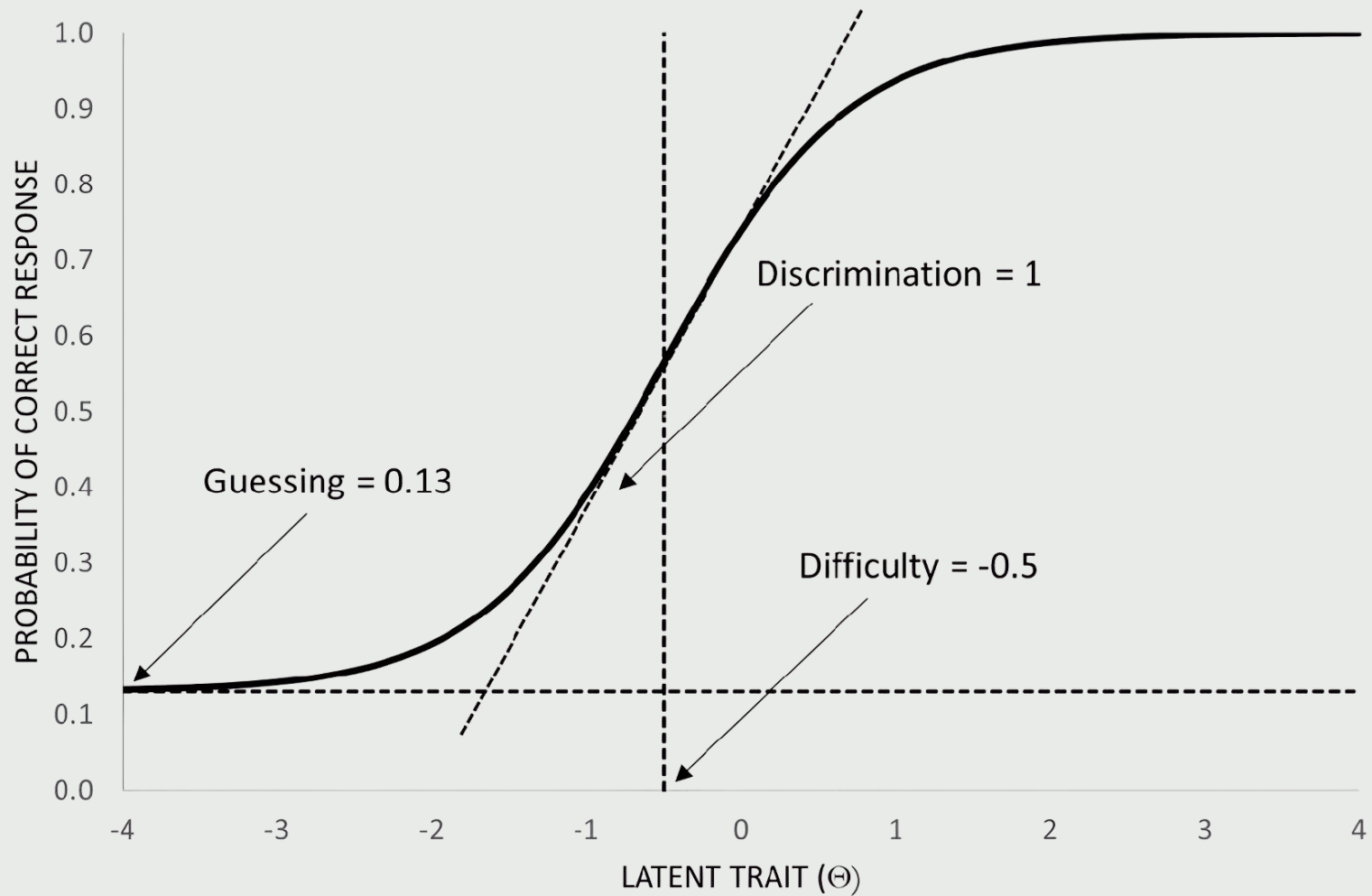


A subject i is characterized by its ability θ_i



An item j is characterized by its difficulty b_j , discrimination a_j , guessing c_j , inattention d_j

Item Response Theory explains the **relationship between an individual's ability and their observed performance** (Columbia University Mailman School of Public Health). Formally, the probability of correctly answering an item is a **function of the person's ability** and visualized as the **item response curve**.



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Glossary

i	examinee i	j	item j	$z = \{\theta, a, b, c, d\}$	a collection of model and item parameters	D	dataset
θ_i	ability of i-th examinee	a_j	discrimination of item j	$\mathbf{r}_{ij} \in \{0, 1\}$	the response of the i-th model to the j-th item	A, B	scaling coefficients
N	number of examinees	b_j	difficulty of item j	$y_{ij} \in \{0, 1\}$	the i-th model correctly (1) or incorrectly (0) responds to the j-th item		
$\mathcal{M}$	the set of (language) models	c_j	probability of guessing for item j				
m_i	i-th model of $\mathcal{M}$	d_j	probability of inattention for item j				
		x_j	the j-th prompt/model input				
		ℓ_j	the j-th ground truth label				
		M	number of items				

In NLP

how it started



how its going



John P. Lalor

“Pioneer of IRT in NLP”

Provided empirical evidence that Item Response Theory *could* be applied to LLMs and their benchmarks

Assistant Professor of IT, Analytics, and Operations at the University of Notre Dame’s Mendoza College of Business

Ph.D., University of Massachusetts-Amherst



Building an Evaluation Scale using Item Response Theory

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¹ University of Massachusetts, MA, USA

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Lalor et al (2016)

- **introduced IRT as a framework for evaluating NLP systems**, challenging the assumption that all test items are equally difficult and discriminative.
- Used Amazon Mechanical Turk workers to annotate a sample of the Stanford Natural Language Inference dataset Recognizing Textual Entailment task
 - Example
 - Premise: Two women are hugging each other.
 - Entailment: Two women are showing affection.
 - The use of 5GS (easier) and 4GS (more difficult) datasets
- fitted a (multidimensional) IRT model to human response data and **used the resulting human ability scale** to compare the performance of NLP systems with that of the human population.

Item Set	Theta Score	Percentile	Test Acc.
5GS			
Entailment	-0.133	44.83%	96.5%
Contradiction	1.539	93.82%	87.9%
Neutral	0.423	66.28%	88%
4GS			
Contradiction	1.777	96.25%	78.9%
Neutral	0.441	67%	83%

Table 5: Theta scores and area under curve percentiles for LSTM trained on SNLI and tested on GS_{IRT} (IRT-based gold standard evaluation scale). Compared to AMT users, the system performed well above average for contradiction items compared to human performance, and performed around average for entailment and neutral items.

Understanding Deep Learning Performance through an Examination of Test Set Difficulty: A Psychometric Case Study

John P. Lalor^{1*}, Hao Wu², Tsendsuren Munkhdalai³, Hong Yu^{1,4}

¹College of Information and Computer Sciences, University of Massachusetts, Amherst

²Department of Psychology and Human Development, Vanderbilt University

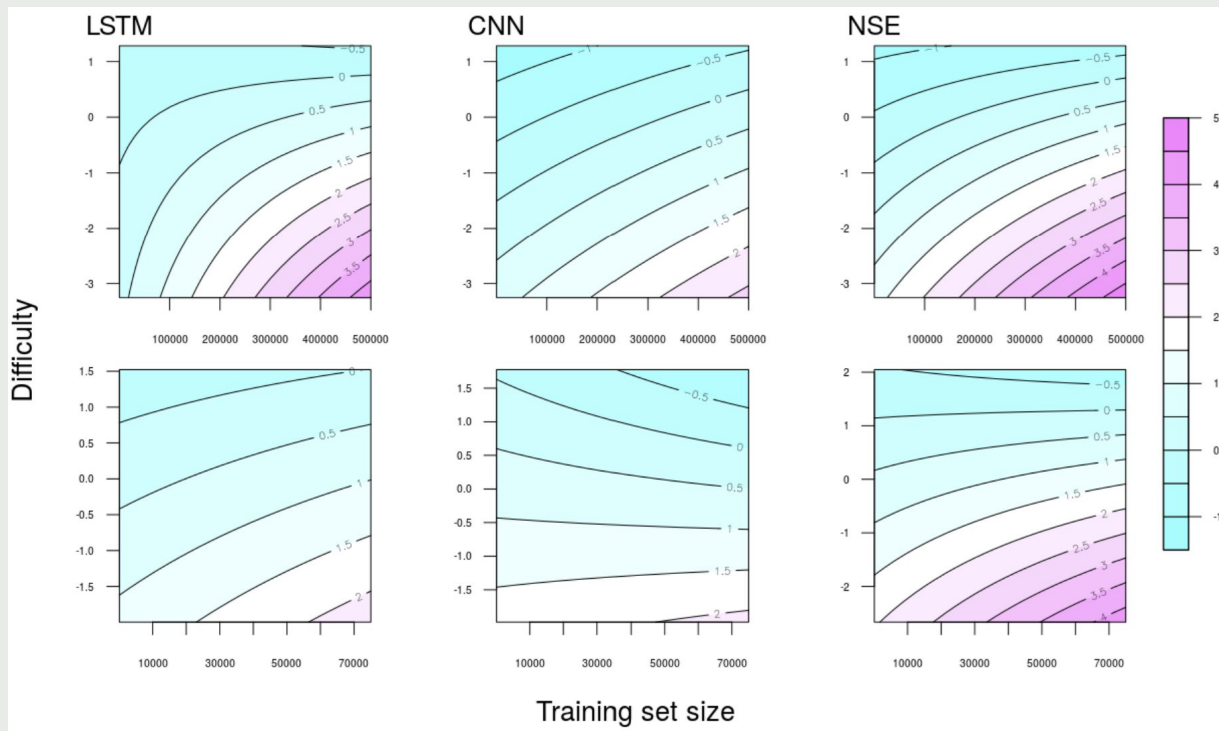
³Microsoft Research, Montréal, Québec

⁴Department of Computer Science, University of Massachusetts, Lowell

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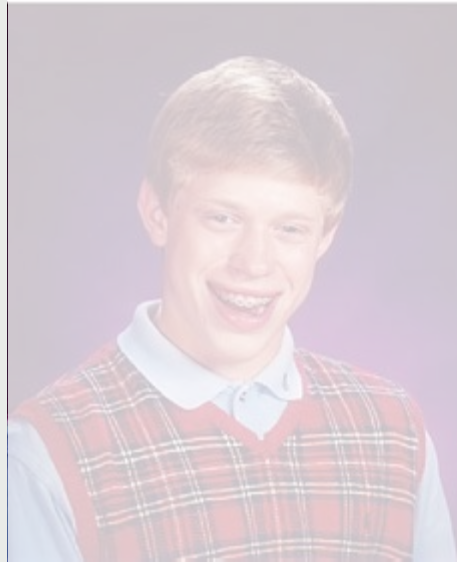
Lalor et al (2020)

- experimented on Natural Language Inference and Sentiment Analysis to show that the **likelihood of answering a question correctly is impacted by the question difficulty**.
- Logistic regression models used training set size and item difficulty as parameters to predict whether a deep neural network would label an item correctly.
- Neural model performance starts to look more human; as DNNs are trained with more data, **“easy” examples are learned more quickly than hard examples**.



Contour plots showing log-odds of labeling an item correctly for NLI (top row) and SA (bottom row) as a function of training set size (x-axis) and item difficulty (y-axis). Blue indicates low log-odds of labeling an item correctly, and pink indicates high log-odds of labeling an item correctly. The contour colors are consistent across plots and log-odds values are shown in the legend on the right.

how it started



how its going



Diagnose benchmark quality

Vania et al (2021) used IRT to answer the question: what kind of datasets are **still effective at discriminating among strong models**? At the time of publication, Quoref, HellaSwag, and MC-TACO were found to best suited for distinguishing among state-of-the-art models.

Rodriguez et al (2021) proposed **DAD (Difficulty and Ability Leaderboards)** that explicitly model both task and submissions jointly. DAD can guide **annotation**, detect annotation **error**, and **partition** evaluation data.

Oliveira and Prudêncio (2022) used IRT to **diagnose what made synthesized speech difficult**: the sentence, locale, and service used to generate speech.

Suppose we collect responses from N models to M benchmark items into an $N \times M$ response matrix. We then train a 4PL IRT model on these responses using an optimization strategy of our choice.

We attain item parameters a_j, b_j, c_j, d_j for M items and model parameters θ_i for N models. We average them respectively to get:

$$\bar{a}, \bar{b}, \bar{c}, \bar{d}, \bar{\theta}$$

Based on these average item and model parameters estimates, how do we know if we have reached benchmark saturation?

High	$\bar{a},$	$\bar{b},$	$\bar{c},$	$\bar{d},$	$\bar{\theta}$
Medium	$\bar{a},$	$\bar{b},$	$\bar{c},$	$\bar{d},$	$\bar{\theta}$
Low	$\bar{a},$	$\bar{b},$	$\bar{c},$	$\bar{d},$	$\bar{\theta}$

Select informative examples and rank systems

Fluid Benchmarking (Hofmann et al, 2025) estimates an item response model based on existing LM evaluation results and uses the inferred quantities to select evaluation items **dynamically**.

metaBench (Kipnis et al, 2025) uses cross-validated score reconstruction to establish what benchmark reduction is feasible and fits an IRT model to those items. Items are then selected based on difficulty and discrimination.

Jiang et al (2026) uses **Fisher information** F to select a micro-benchmark, a smaller version of a benchmark

Zouhar et al (2026) selects items with the **highest difficulty** $\times$ **discrimination product**.

Worthy Opponents

*Randomly sampled
dataset*

$$D_{\text{random}} \sim \text{Unif}(\{D' \subseteq D_{\text{full}} \mid |D'| = m\})$$

*Randomly stratified
datasets*

$$D_{\text{stratified random}} = \bigcup_{k=1}^t D'_k \text{ where } t \text{ is the number of subtasks}$$

$$D_{\text{full}} = \cup_{k=1}^t D_k, \text{ and}$$

$$D'_k \sim \text{Unif}(\{D'_k \subseteq D_k \mid |D'_k| = \lfloor m/t \rfloor\})$$

Microbenchmarking: techniques for identifying small evaluation sets that faithfully represent model behavior on a large dataset.

$$\sum_{j \in D_{\text{micro}}} w_j y_{ij} \approx \frac{1}{|D_{\text{full}}|} \sum_{j \in D_{\text{full}}} y_{ij}$$

How can researchers model the information provided by a specific benchmark item in relation to a model's performance?

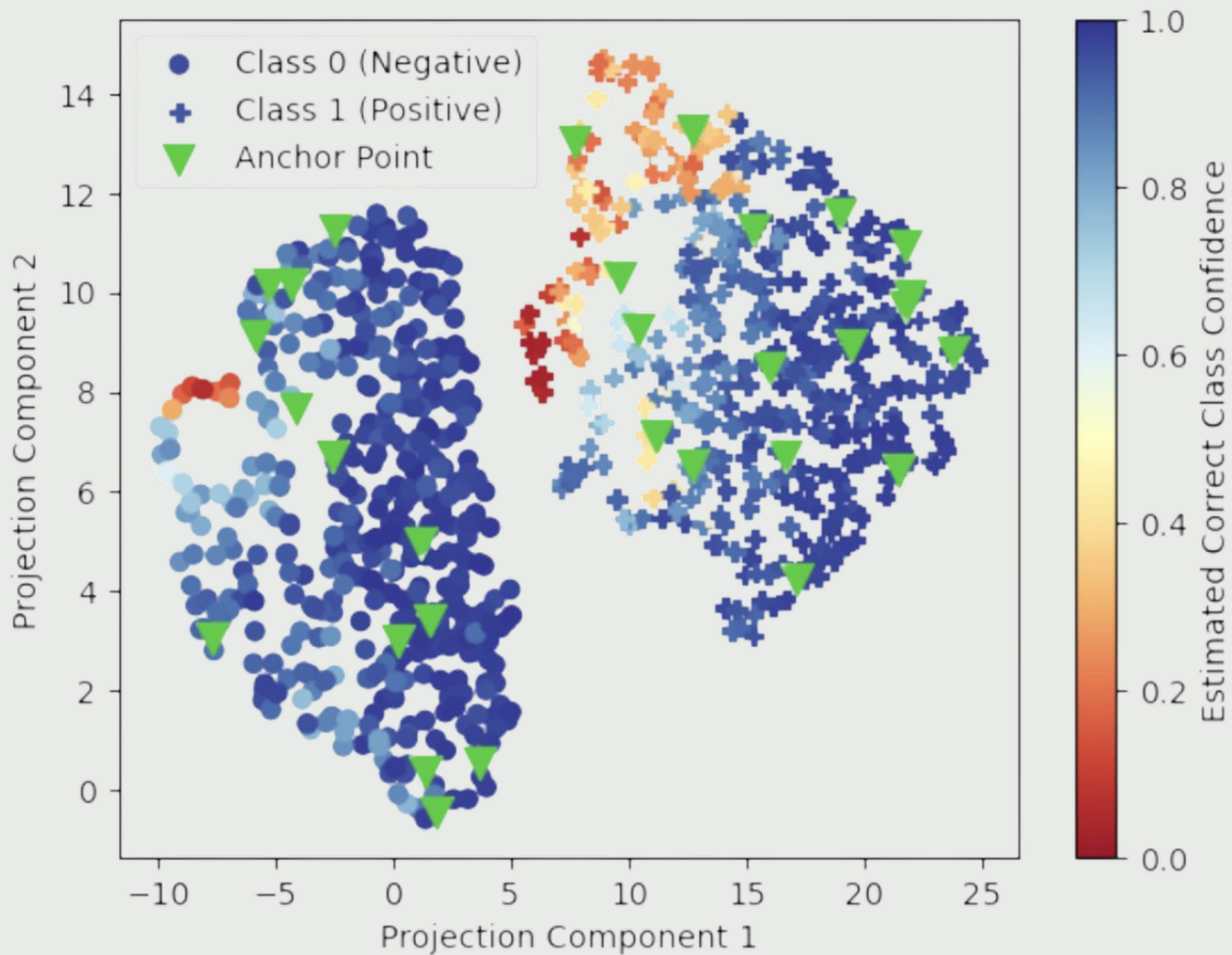
Anchor Points (not IRT)

The predictions of existing source models on a given dataset are telling about predictions of a new target model. Solved using Partitioning Around Medoids algorithm.

$$\text{CORR}_S(j, k) = \text{PearsonCorrelation}(\mathbf{logit}(\mathbb{P}[\ell_j \mid x_j]), \mathbf{logit}(\mathbb{P}[\ell_k \mid x_k]))$$

$$D_{\text{Anchor Points}}^* = \underset{\substack{D_{\text{Anchor Points}} \subset D_{\text{full}} \\ |D_{\text{Anchor Points}}| = B}}{\text{argmin}} \sum_{j \in D_{\text{full}}} \min_{k \in D_{\text{Anchor Points}}} [1 - \text{CORR}_S(j, k)]$$

where B is the number of anchor points



Anchor Points (not IRT)

1. Compute pairwise correlations between all points using CORR_S creating a $|D_{\text{full}}| \times |D_{\text{full}}|$ correlation matrix C
2. Represent distance matrix $Z=1-C$ as positive, monotonically decreasing function of correlation
3. Cast the points to a high dimensional continuous space using Multi-Dimensional Scaling on Z and project the space to two components with U-MAP

In Item Response Theory (IRT), an anchor point refers to a shared reference baseline used to link, equate, or calibrate different test forms onto a single, standardized measurement scale.

tinyBenchmarks

correctness-based representation

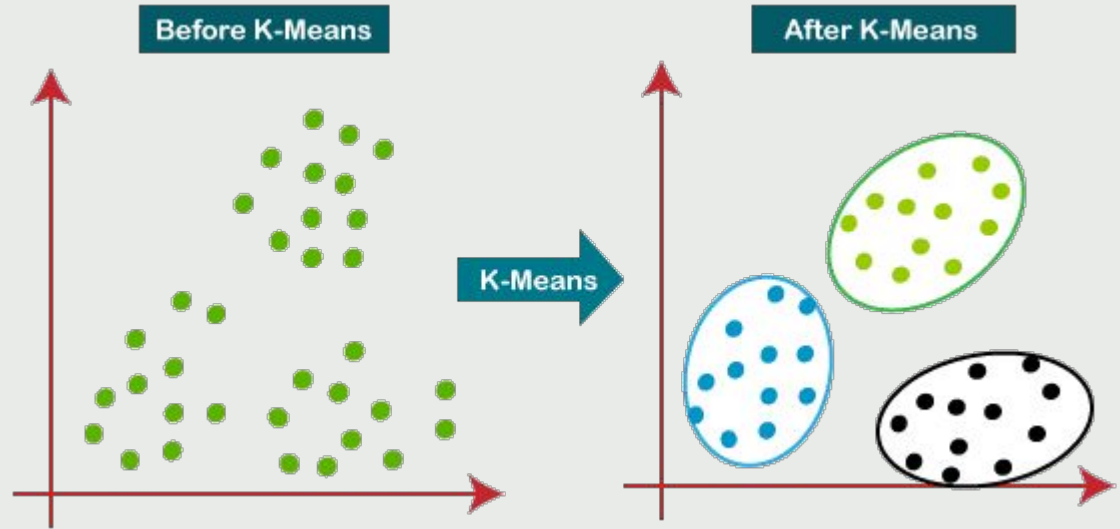
model ↓ item →		0	1	2	3	4
0	1	1	1	1	1	1
1	0	0	0	0	0	0
2	1	0	1	0	1	1
3	0	1	0	1	0	0
4	1	1	1	0	1	1

IRT-based representation

IRT ↓		0	1	2	3	4
item →						
a_j		0.02	0.32	0.51	0.12	0.16
b_j		0.08	0.25	0.10	0.64	0.04

tinyBenchmarks

1. Runs k-means clustering on a representation of a benchmark item
2. After the k centroids are obtained, find the closest example to each centroid and each of these points will compose D_{micro}
3. Set w_i as the fraction of points in D_{full} assigned to the i th cluster



tinyBenchmarks

correctness-based representation	IRT-based representation
Suffers from distribution shifts since correctness patterns can vary	Creates meaningful representations for each benchmark item based on their difficulties and abilities
Suffers from the curse of dimensionality	Solves the dimensionality problem since the representation is 2-dimensional

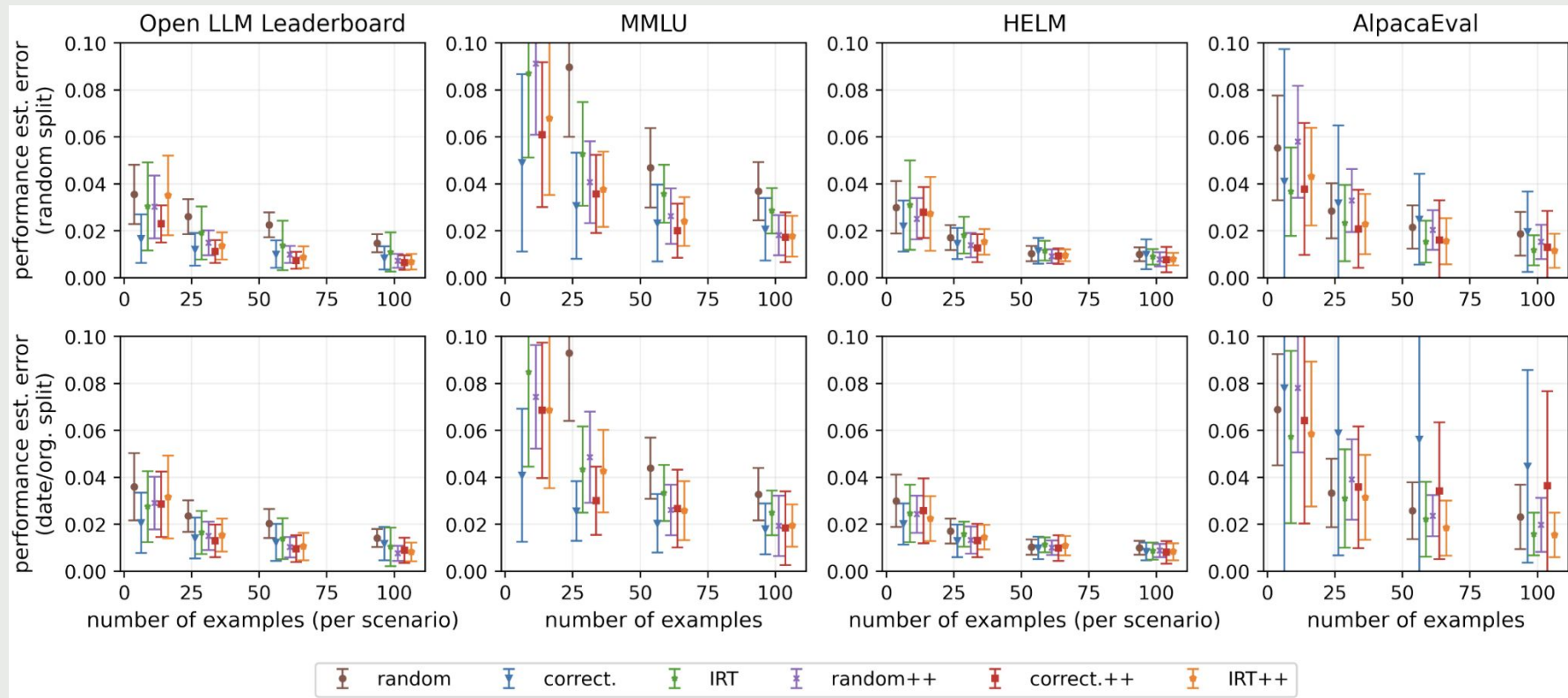


Figure 3: Performance estimation error per benchmark (columns) tested on random (top row) and recent (bottom row LLMs) for increasing number of evaluation examples. **A hundred examples per scenario** is sufficient to achieve $\approx 2\%$ average performance estimation error across benchmarks and evaluated LLMs. Vanilla methods (no ++) take the selected questions and calculate their weighted average. The paper introduces a gp-estimator (has ++) as a convex combination of the raw weighted average and an estimate whose weight depends on **estimated IRT bias and sampling variance**.

Fisher information is way of measuring the amount of information that an observable random variable G carries about an unknown parameter h upon which the probability of G depends

In the case of IRT in NLP, G is whether model i gets item j correct. Unknown parameter h is composed of model and item parameters.

$$\hat{\mathcal{F}}_j(\hat{\theta}_i) = \hat{a}_j^2 \cdot \frac{(\mathbb{P}[\mathbf{r}_{ij} = 1 \|\hat{\theta}_i, \hat{a}_j, \hat{b}_j, \hat{c}_j, \hat{d}_j])^2}{(1 - \hat{c}_j)^2} \cdot \frac{1 - \mathbb{P}[\mathbf{r}_{ij} = 1 \|\hat{\theta}_i, \hat{a}_j, \hat{b}_j, \hat{c}_j, \hat{d}_j]}{\mathbb{P}[\mathbf{r}_{ij} = 1 \|\hat{\theta}_i, \hat{a}_j, \hat{b}_j, \hat{c}_j, \hat{d}_j]}$$

Jiang et al (2026): Select the top 10% of items with the largest Fisher information averaged over ability to create a smaller benchmark

Fluid Benchmarking

$$\text{EVALUATE}(m_i, D) = \underbrace{\text{AGGREGATE}_{q_j \in \text{SELECT}(D)}}_{\text{benchmark-level aggregation}} \underbrace{(\text{SCORE}(m_i, q_j))}_{\text{item-level scoring}}$$

SELECT is a selection function that determines the set of evaluation items

SCORE is a scoring function that quantifies LM performance on each item in the evaluation set

If $\text{SCORE} \in \{0, 1\}$ is a binary function, AGGREGATE returns the mean of the item-level scores, and $\text{SELECT}(D) = D$,

we can recover the standard accuracy metric, which we denote as $\text{ACCURACY}(m_i, D)$.

Fluid Benchmarking: changing AGGREGATE

$$\text{SCORE}(m_i, q_j) = y_{ij}$$

$$\text{ABILITY}(m_i, D) = \text{AGGREGATE}(m_i, D) = \hat{\theta}_i$$

Fluid Benchmarking: changing SELECT

$$I(\theta_i, a_j, b_j) = a_j^2 \cdot \mathbb{P}[\mathbf{r}_{ij} = 1 \mid \theta, a_j, b_j, c_j, d_j] \cdot (1 - \mathbb{P}[\mathbf{r}_{ij} = 1 \mid \theta, a_j, b_j, c_j, d_j])$$

Fisher information $I(\theta_i^t, a_j, b_j)$

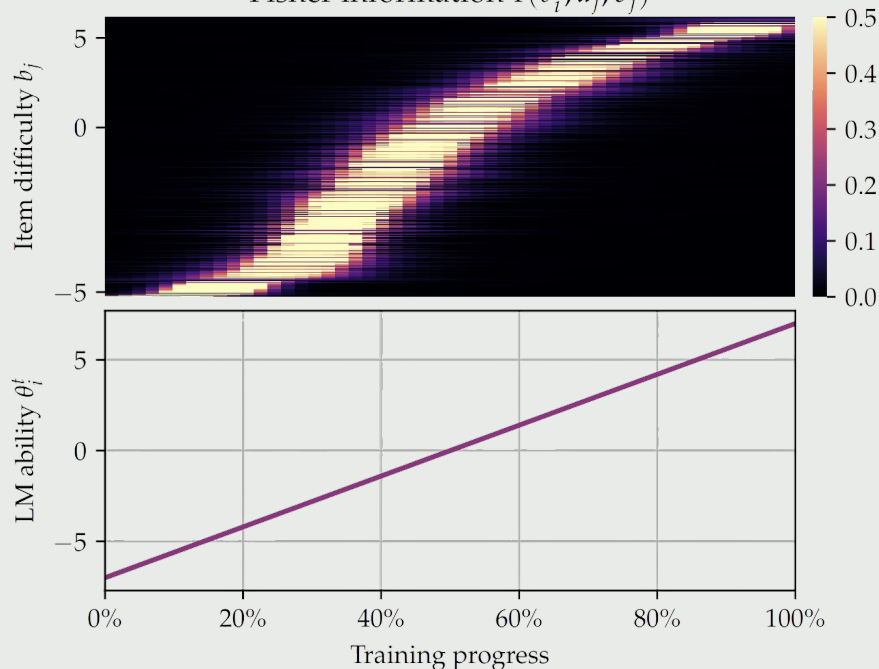


Figure 2: Fisher information of HellaSwag items as a function of training progress. Lower panel: simulated trajectory of language model ability, which evolves linearly from $\theta_i^1 = -7$ to $\theta_i^{50} = +7$; upper panel: Fisher information of HellaSwag items. The HellaSwag items with highest Fisher information change drastically. The items with the highest Fisher information changes from very easy items at the beginning of training to very difficult items at the end of training.

Fluid Benchmarking: changing SELECT

$$D_i^*(0) = \emptyset, \quad D_i^*(t) = D_i^*(t-1) \cup \left\{ \arg \max_{q_j \in Q \setminus D_i^*(t-1)} I(\text{ABILITY}(m_i, D_i^*(t-1)), a_j, b_j) \right\}$$

1. Iteratively select the item from D with the highest Fisher information given the current ability estimate
2. Repeat procedure until the total number of administered items has reached the budgeted size

Arguments for (9/17) and against (9/18)
using Item Response Theory in NLP

Should we be doubtful?

Literature Review

AI benchmark data often **departs from the data regime of human testing**, for which standard IRT estimation tools were originally developed: (AI) benchmarks typically involve **fewer evaluated models, far more items**, and **capability distributions may be skewed, clustered or multimodal** (Jiang et al, 2026).

Model evaluation suffers from high variance when the benchmark has a **small number of examples** (Fisch et al, 2026).

Benchmark prediction sharply declines when **new models have higher accuracy** than previously seen models (Zhang et al, 2026).

Minimum Detectable Ability Difference (Yauney et al, 2026) investigates how well a microbenchmark can rank two models as a **function of their performance difference on the full benchmark**. Researchers found that (i) microbenchmarks cannot rank models as consistently as the **full benchmarks they replace** and (ii) could not outrank models more consistently than selecting a **random subset of data points**.

Can We Trust Item Response Theory for AI Evaluation?

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ALL response data

IRT Model

Variational
Inference

Ground Truth
Model, Item
Parameters

$\subset$ ALL response data

IRT Model

Optimization

Predicted Model,
Item Parameters

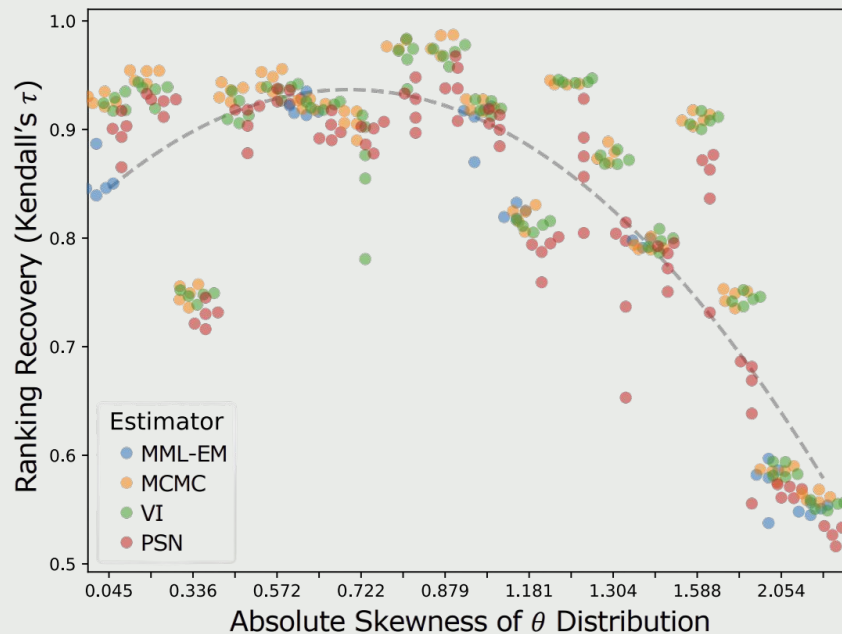


Figure 3: Relationship between capability distribution skewness and model ranking recovery. Each point represents one simulation condition; sample sizes are pooled without visual distinction. The absolute skewness varies across benchmark-IRT model combinations. The dashed line shows a quadratic fit to all points.

Skewness is a statistical measure that describes the lack of symmetry in a data distribution

Normality assumption: The dominant factor affecting ranking recovery is the **skewness of the true capability distribution**. The ranking recovery exceeded 0.85 for conditions with low skewness ($|\gamma| < 0.5$), but **degraded substantially when skewness increased**, falling below 0.60 for the most skewed conditions ($|\gamma| > 2.0$) (Jiang et al, 2026).

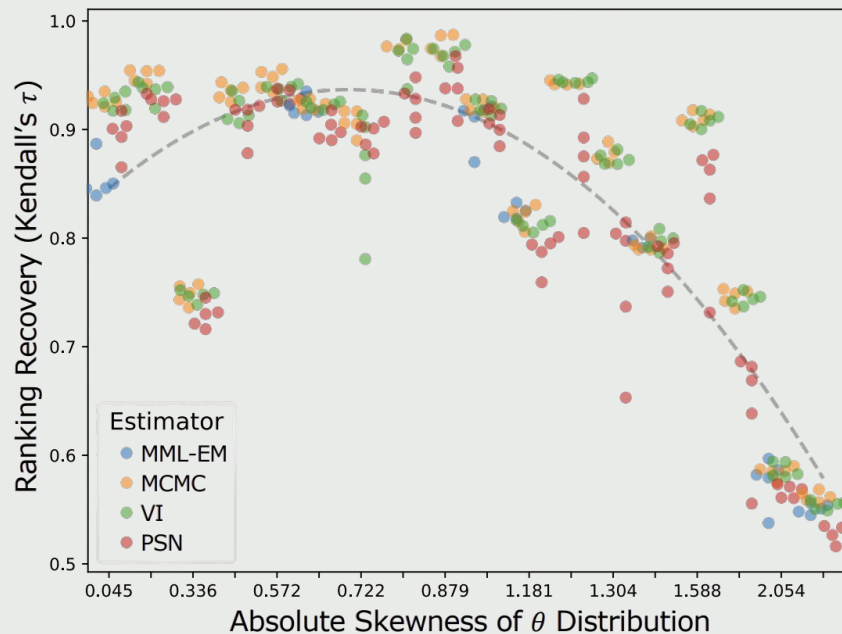


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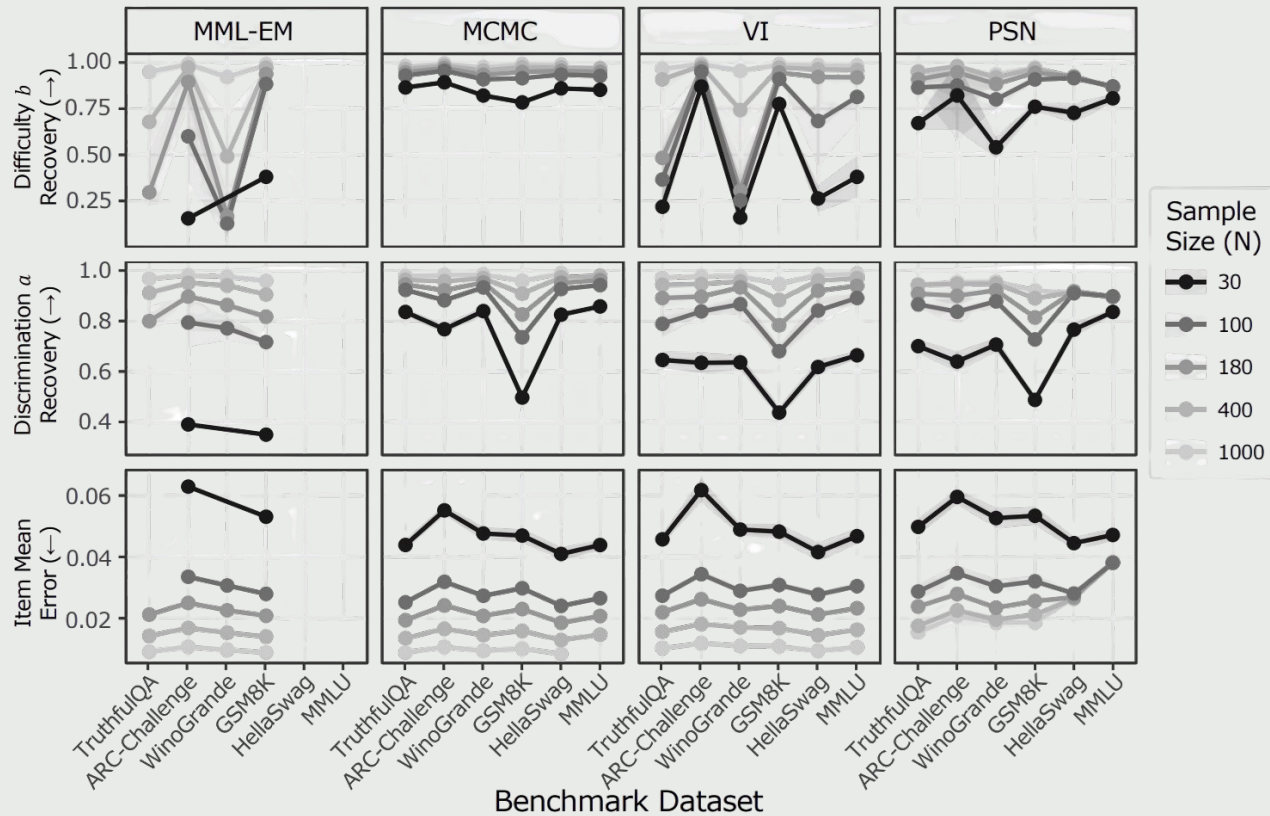


Figure 5: Item parameter recovery under 2PL across estimators, benchmarks, and sample sizes. Higher is better for b and a recovery; lower is better for item mean error. Shaded regions indicate ± 1 standard deviation across 50 replications. Missing points indicate computational infeasibility.

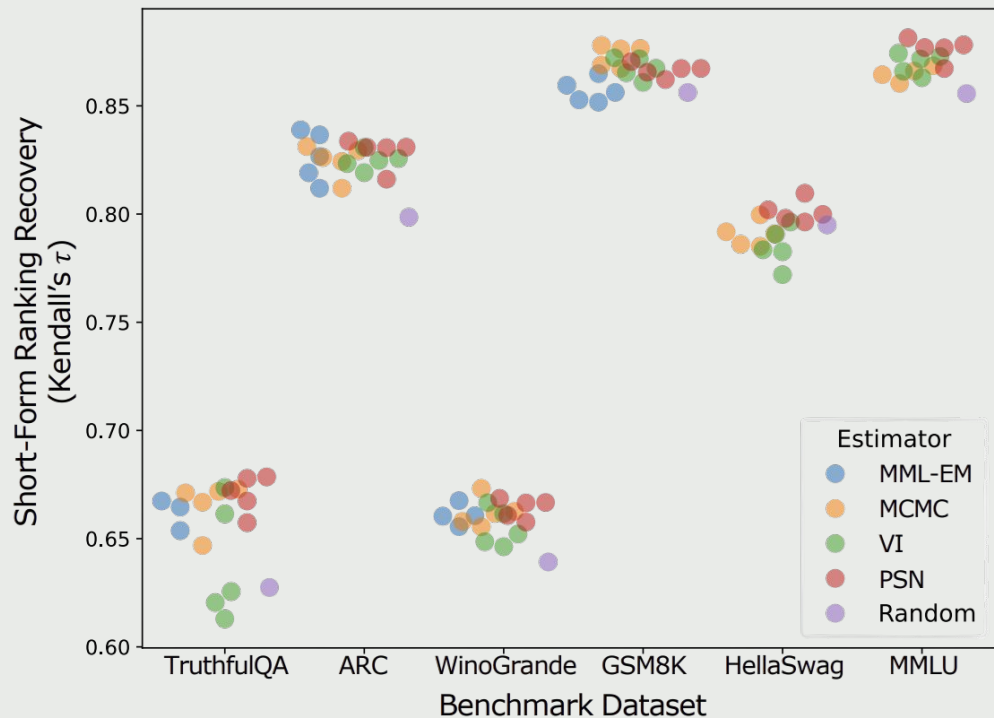


Figure 4: Short-form ranking recovery across benchmarks and estimators (2PL). Each point represents one simulation condition; sample sizes are pooled without visual distinction.

IRT-based benchmark compression **consistently outperformed** the random baseline across benchmarks, though the margin was **modest**. Neither sample size nor the choice of estimates had a strong effect on short-form ranking recovery. Short-form ranking recovery showed no monotonic relationship with the skewness of capability distribution.



≠



Too Many Meta-Evaluation Metrics: Benchmark Size

$$\text{efficiency} = |D|$$

e.g. $|D_{\text{tinyBenchmarks}}| = 100 = \text{efficiency}$

Meta-Evaluation Metric Counter
1

Too Many Meta-Evaluation Metrics: failure rate

$$\text{failure rate} = \frac{\text{number of replications that failed to produce parameter estimates}}{\text{total number of unique runs} \times \text{number of replications per run}}$$

e.g. total number of failed replicates = 101

total number of unique runs = 10

number of replications per run = 50

$$\text{failure rate} = \frac{101}{10 \times 50} = 20.2\%$$

Meta-Evaluation Metric Counter
2

Too Many Meta-Evaluation Metrics: runtime

$$\text{per-replication runtime} = \frac{1}{\# \text{ of replications}} \sum_{r=1}^{\# \text{ of replications}} \text{runtime } r$$

e.g. runtimes = [5, 10, 15, 20, 60] (in seconds)

$$\text{per-replication runtime} = \frac{5 + 10 + 15 + 20 + 60}{5} = 22 \text{ seconds}$$

Meta-Evaluation Metric Counter
3

Aside: Ranking

Rank	Model	Score
1	GPT-6 Astra	92.4
2	GPT-5.6 Sol	92
3	Claude Opus 5	91.6
4	Claude Fable 5	91.5
5	Gemini-3.1-Pro	91
	...	

Snapshot of MMLU-Pro leaderboard as of September 11th, 2026 7:49 PM Central

Too Many Meta-Evaluation Metrics: ranking recovery

Kendall's $\tau = \frac{C - D}{C + D}$ where C represents concordant pairs and D discordant pairs

Language Model	Ground Truth Ranking	Predicted Ranking	Concordant Pairs	Discordant Pairs
GPT-4o	1	2	2	1
Claude 3.5	2	1	2	0
Gemini 1.5 Pro	3	4	0	1
Mistral Large	4	3	0	0

Kendall's $\tau = \frac{4 - 2}{4 + 2} \approx 0.333$

Meta-Evaluation
Metric Counter

4

Too Many Meta-Evaluation Metrics: ranking recovery

$$\rho_{\text{Spearman}} = \rho_{\text{Pearson}} = \frac{\text{cov}(\text{rank } \mathcal{X}, \text{rank } \mathcal{Y})}{\sigma_{\text{rank } \mathcal{X}} \sigma_{\text{rank } \mathcal{Y}}}$$

Language Model	Ground Truth Ranking	Predicted Ranking	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$	$(X - \bar{X})^2$	$(Y - \bar{Y})^2$
GPT-4o	1	2	-1.5	-0.5	0.75	2.25	0.25
Claude 3.5	2	1	-0.5	-1.5	0.75	0.25	2.25
Gemini 1.5 Pro	3	4	+0.5	+1.5	0.75	0.25	2.25
Mistral Large	4	3	+1.5	+0.5	0.75	2.25	0.25
SUM					3.00	5.00	5.00

Too Many Meta-Evaluation Metrics: ranking recovery

$$\rho_{\text{Spearman}} = \rho_{\text{Pearson}} = \frac{\text{cov}(\text{rank } \mathcal{X}, \text{rank } \mathcal{Y})}{\sigma_{\text{rank } \mathcal{X}} \sigma_{\text{rank } \mathcal{Y}}}$$

$$\text{cov}(\text{rank } \mathcal{X}, \text{rank } \mathcal{Y}) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n} = \frac{3.00}{4} = 0.75$$

$$\sigma_x = \sqrt{\frac{5.00}{4}} = \sqrt{1.25} \approx 1.118, \quad \sigma_y = \sqrt{\frac{5.00}{4}} \approx 1.118$$

$$\rho_{\text{Spearman}} = \frac{0.75}{1.118 \times 1.118} = \frac{0.75}{1.25} = 0.60$$

Meta-Evaluation
Metric Counter

5

Too Many Meta-Evaluation Metrics: ranking recovery

$$\text{Top1}(\mathcal{X}, \mathcal{Y}) = \mathbb{1} \left[\underset{m \in \mathcal{M}}{\operatorname{argmax}} \operatorname{score}_x(m) = \underset{m \in \mathcal{M}}{\operatorname{argmax}} \operatorname{score}_y(m) \right]$$

$$\text{e.g. } \text{Top1}(\mathcal{X}, \mathcal{Y}) = \mathbb{1} \left[\underbrace{\text{GPT-4o}}_{\text{Ground-truth}} = \underbrace{\text{Claude 3.5}}_{\text{Predicted ranking}} \right] = 0$$

Meta-Evaluation Metric Counter
6

Too Many Meta-Evaluation Metrics: ranking recovery

Validity compute the distance between an LMs' predicted ranks on two different benchmarks. To test how well estimated performance on one benchmark predicts performance on a different benchmark that targets the same capability

$$\text{validity} = \sum_{m_i \in M} |\text{rank}_{\text{benchmark } x}(m_i) - \text{rank}_{\text{benchmark } y}(m_i)|$$

Too Many Meta-Evaluation Metrics: ranking recovery

Language Model	Ground Truth Ranking	Predicted Ranking	$ \text{rank}_{\text{benchmark } x}(m_i) - \text{rank}_{\text{benchmark } y}(m_i) $
GPT-4o	1	2	+1
Claude 3.5	2	1	+1
Gemini 1.5 Pro	3	4	+1
Mistral Large	4	3	+1
VALIDITY			+4

Meta-Evaluation
Metric Counter

7

Too Many Meta-Evaluation Metrics: ranking recovery

Yauney et al (2026) defined agreement for each bucket $B \in \mathcal{B}$

$$\text{agreement}(D_{\text{micro}}, D_{\text{full}}, B) = \mathbb{P}_{m_1, m_2 \in \mathcal{M}} (\Delta_{D_{\text{micro}}}(m_1, m_2) > 0 \mid \Delta_{D_{\text{full}}}(m_1, m_2) \in B)$$

where $\Delta_D(m_1, m_2) = \text{score}_D(m_1) - \text{score}_D(m_2)$

$\mathcal{B}$ is a set of ordered buckets of pairwise differences

w.l.o.g. assume that the model with higher performance on the full benchmark is m_1

Too Many Meta-Evaluation Metrics: ranking recovery

$$\text{MDAD}(D_{\text{micro}}, D_{\text{full}}) = \arg \min_{\substack{B \in \mathcal{B} \\ \text{centroid}(B)}} \{\text{agreement}(D_{\text{micro}}, D_{\text{full}}, B)\} \quad \text{s.t.} \quad \mathbb{P}[\dots] \geq 0.8$$

MDAD seeks to answer the question: what is the lowest performance difference on the full benchmark at which pairwise model rankings on a microbenchmark are consistent with those on the full benchmark?

Meta-Evaluation Metric Counter
8

Too Many Meta-Evaluation Metrics: ranking recovery

Pair(m_1, m_2)	$\Delta_{D_{\text{full}}}$	$\Delta_{D_{\text{micro}}}$	$\Delta_{D_{\text{micro}}} > 0$
GPT-4o vs Claude 3.5	0.90-0.85=0.05	0.85-0.88 = -0.03	FALSE
GPT-4o vs Gemini	0.90-0.75=0.15	0.85-0.70=+0.15	TRUE
GPT-4o vs Mistral	0.90-0.70=0.20	0.85-0.73=+0.12	TRUE
...	...	...	..
Claude 3.5 vs Mistral	0.80-0.70=0.15	0.88-0.73=+0.15	TRUE
Gemini vs Mistral	0.75-0.70=0.05	0.70-0.73=-0.03	FALSE

Too Many Meta-Evaluation Metrics: ranking recovery

Bucket B	Pairs in Bucket	$P[\Delta_{D_{\text{micro}}} > 0]$	Centroid	satisfies ≥ 0.8
[0.00, 0.07)	GPT-4o vs Claude (0.05), Gemini vs Mistral (0.05)	0/2	0.05	NO
[0.07, 0.13)	Claude vs Gemini (0.10)	1/1=1.00	0.10	YES
[0.13, 0.25)	PT-4o vs Gemini (0.15), GPT-4o vs Mistral (0.20), Claude vs Mistral (0.15)	3/3 = 1.00	0.19	YES

Probability micro-benchmark agrees with full benchmark

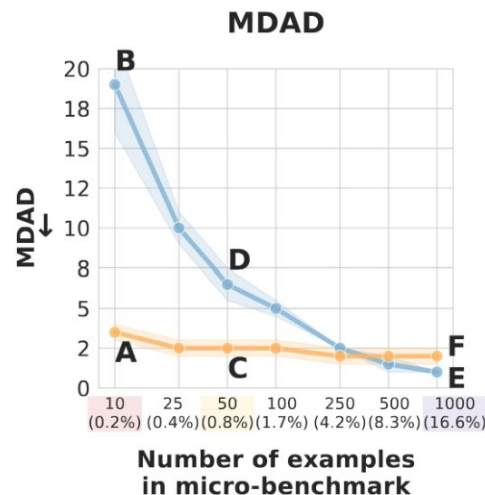
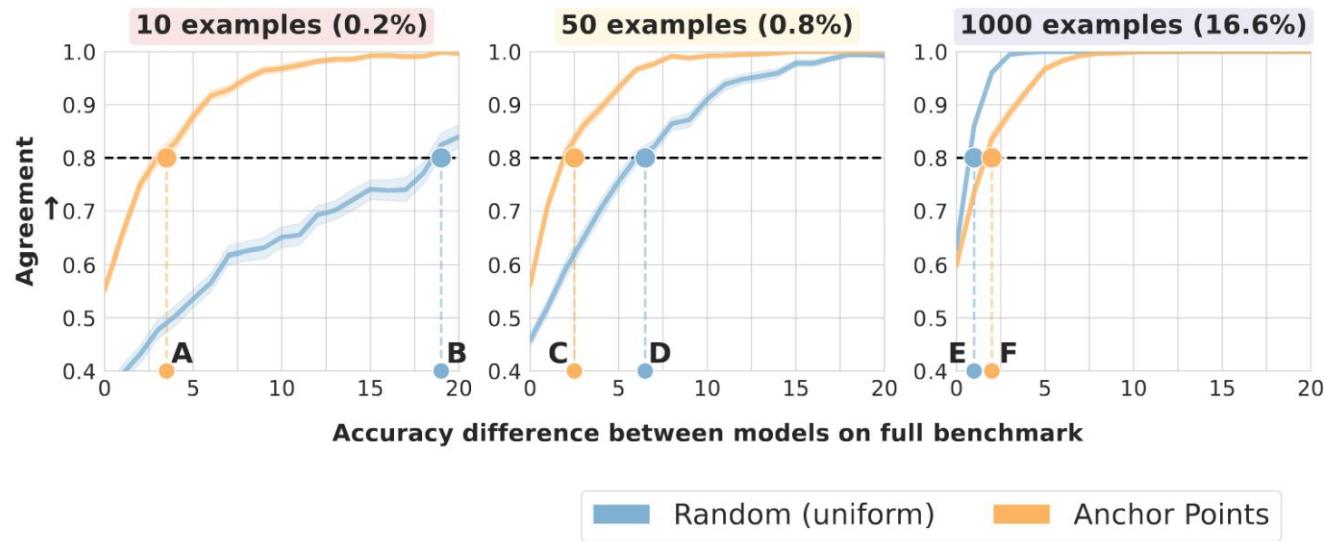


Figure 2: Agreement and MDAD measures on MMLU-Pro for uniform random sampling and AnchorPoints. The three left panels show the probability that a pairwise ranking of models on a smaller benchmark agrees with the full benchmark's ranking, as a function of the accuracy difference between those models on the full benchmark. The rightmost panel summarizes all agreement curves by showing the minimum detectable accuracy difference between models at each (micro)benchmark size, i.e. the accuracy difference at which each curve in the first three panels crosses the 0.8 probability of agreement threshold. For MDAD, lower values are better. Error bars represent 95% bootstrap confidence intervals over 50 trials

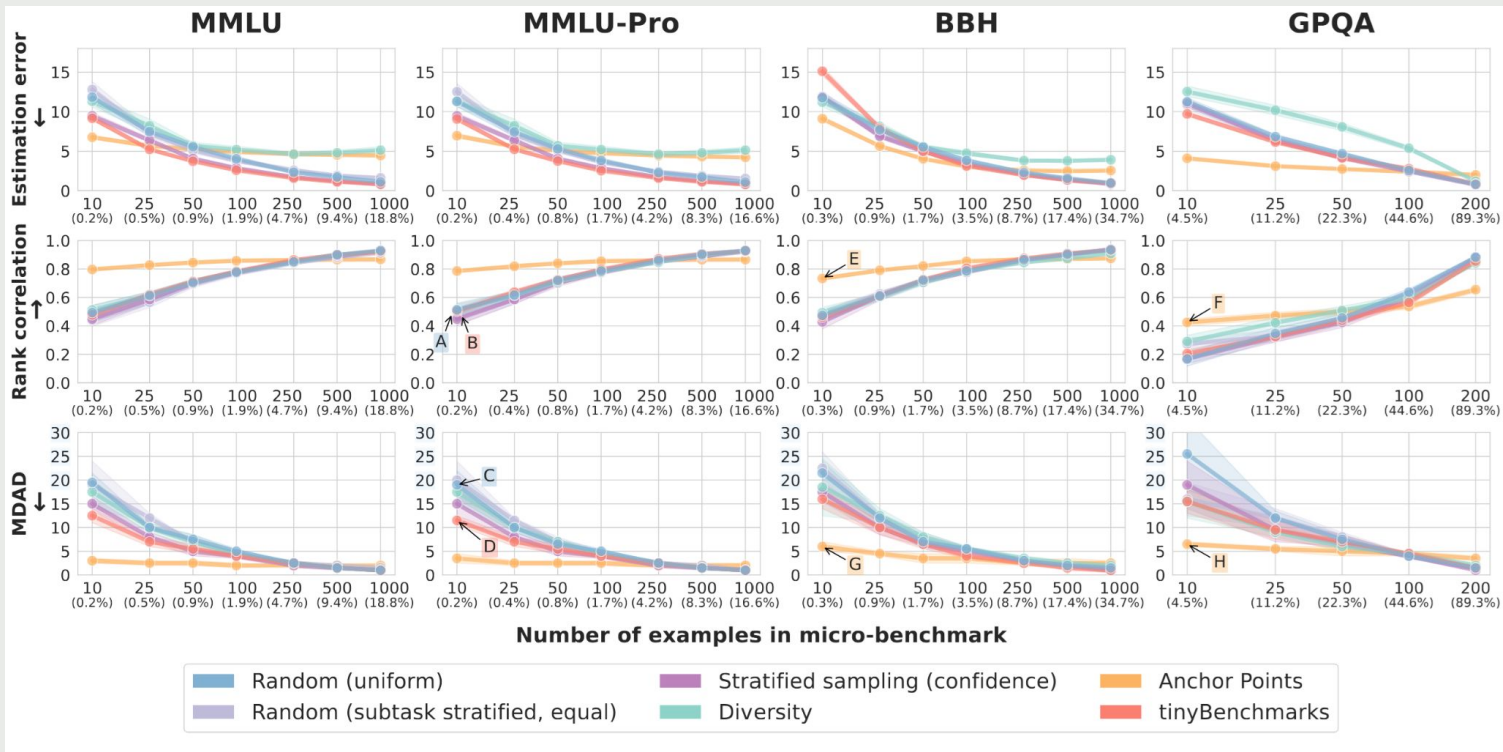


Figure 4: MDAD gives more granular information than mean estimation error and Kendall’s tau rank correlation. AnchorPoints is the only method that consistently outperforms random sampling at small dataset sizes across all metrics. For small benchmarks, all methods struggle to compare models that differ by fewer than 4 points of accuracy on the full benchmark. Error bars represent 95% bootstrap confidence intervals over 50 trials.

Too Many Meta-Evaluation Metrics: score error

$$\text{MAE}(\mathcal{Y}, \mathcal{X}) = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} |\text{score}_{\mathcal{Y}}(m) - \text{score}_{\mathcal{X}}(m)|$$

$$\text{RMSE}(\mathcal{Y}, \mathcal{X}) = \sqrt{\frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} (\text{score}_{\mathcal{Y}}(m) - \text{score}_{\mathcal{X}}(m))^2}$$

Meta-Evaluation Metric Counter
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Too Many Meta-Evaluation Metrics: ranking recovery

Model	$\text{score}_{D_{\text{full}}}$	$\text{score}_{D_{\text{micro}}}$	$\text{score}_{D_{\text{full}}} - \text{score}_{D_{\text{micro}}}$	$ \cdot $	$(\cdot)^2$
GPT-4o	0.90	0.85	-0.05	0.05	0.0025
Claude 3.5	0.85	0.88	+0.03	0.03	0.0009
Gemini 1.5 Pro	0.75	0.70	-0.05	0.05	0.0025
Mistral Large	0.70	0.73	0.03	0.03	0.0009
Sum				0.16	0.0068

$$\text{MAE} = \frac{0.16}{4} = 0.04 \quad \text{RMSE} = \sqrt{\frac{0.0068}{4}} \approx 0.0041$$

Too Many Meta-Evaluation Metrics: item characteristic curve

$$\text{aggregate score error} = \frac{1}{N} \left[\sum_{i=1}^N \frac{1}{\mathcal{M}} \left(\sum_{j=1}^{\mathcal{M}} \left| \underbrace{\mathbb{P}[r_{ij} = 1 \mid \theta_i, a_j, b_j, c_j, d_j]}_{\text{true parameters}} - \underbrace{\mathbb{P}[r_{ij} = 1 \mid \hat{\theta}_i, \hat{a}_j, \hat{b}_j, \hat{c}_j, \hat{d}_j]}_{\text{estimated parameters}} \right| \right) \right]$$

Meta-Evaluation
Metric Counter

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Too Many Meta-Evaluation Metrics: item parameter recovery

$$\text{discrimination recovery} = \frac{1}{M} \sum_{j=1}^M |\hat{a}_j - a_j|$$

$$\text{difficulty recovery} = \frac{1}{M} \sum_{j=1}^M |\hat{b}_j - b_j|$$

$$\text{guessing recovery} = \frac{1}{M} \sum_{j=1}^M |\hat{c}_j - c_j|$$

$$\text{mean item error} = \frac{1}{M} \left[\sum_{j=1}^M \frac{1}{N} \left(\sum_{i=1}^N \left| \underbrace{\mathbb{P}[r_{ij} = 1 \mid \theta_i, a_j, b_j, c_j, d_j]}_{\text{true parameters}} - \underbrace{\mathbb{P}[r_{ij} \mid \hat{\theta}_i, \hat{a}_j, \hat{b}_j, \hat{c}_j, \hat{d}_j]}_{\text{estimated parameters}} \right| \right) \right]$$

Meta-Evaluation
Metric Counter

15

Too Many Meta-Evaluation Metrics: Adaptive Testing

$$\text{TV}(m_i, Q) = \frac{\# \text{ of checkpoints}}{\# \text{ of checkpoints} - 1} \times \frac{\sum_{t=1}^{\# \text{ of checkpoints} - 1} |\text{EVALUATE}(m_i^{t+1}, D) - \text{EVALUATE}(m_i^t, D)|}{|\text{EVALUATE}(m_i^n, D) - \text{EVALUATE}(m_i^1, D)|}$$

$$\text{saturation} = |\rho_{\text{spearman}}(\{t, \text{predicted performance}\}_t = 1 : \# \text{ of checkpoints})|$$

Meta-Evaluation
Metric Counter

17

An open question
Can IRT be applied to multilingual
evaluation? If so, what steps do
researchers need to take?

Many thanks to Dr. Karen Livescu and Vilém Zouhar for sourcing much of the presented literature.